Calculator & Test Strategy

ANSWER KEY

Choosing between graphing and algebra

1 A system is given by $y = x^2 - 4x + 3$ and $y = 2x - 6$. Explain how graphing both functions on a calculator can quickly reveal the number of solutions, then find the solutions algebraically.

Graphing both functions shows how many times the parabola and line intersect — the number of intersection points equals the number of real solutions, which can be visually confirmed before doing algebra. Algebraically: $x^2 - 4x + 3 = 2x - 6$, so $x^2 - 6x + 9 = 0$, factoring to $(x - 3)^2 = 0$, giving the single repeated solution $x = 3$ (matching a graph that shows the line tangent to the parabola at one point).

Using a calculator to check algebraic work

2 A student solves $3x - 7 = 2x + 5$ and gets $x = 12$. Verify this solution by substitution, and explain why checking answers this way is a valuable test-taking strategy.

Substitute $x = 12$: LHS $= 3(12) - 7 = 36 - 7 = 29$. RHS $= 2(12) + 5 = 24 + 5 = 29$. Since both sides match, $x = 12$ is correct. Checking by substitution is valuable because it catches arithmetic slips made during solving, without needing to redo the entire algebraic process — a quick, reliable way to build confidence before moving to the next problem.

Estimating and eliminating unreasonable answer choices

3 A multiple-choice question asks for the value of $\sqrt{50}$, with choices A) 5, B) 7.07, C) 25, D) 100. Without a calculator, use estimation to eliminate unreasonable choices, then find the exact simplified value.

Since $7^2 = 49$ and $8^2 = 64$, $\sqrt{50}$ must be just above 7, immediately eliminating A (5, too small), C (25, since 25^2 is way too large), and D (100, absurdly large). This leaves B (7.07) as reasonable. Exact simplified form: $\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2} \approx 7.07$, confirming B.

Efficient use of the calculator for systems of equations

4 Explain two different calculator-based strategies (other than the built-in equation solver) for solving the system $2x + 3y = 12$ and $x - y = 1$, and then solve it algebraically to confirm.

Strategy 1: graph both lines and find their intersection point visually. Strategy 2: rewrite both equations in $y = mx + b$ form and use a table function to find where the y-values match for the same x. Algebraically: from the second equation, $x = y + 1$. Substitute: $2(y + 1) + 3y = 12$, so $2y + 2 + 3y = 12$, giving $5y = 10$, $y = 2$. Then $x = 2 + 1 = 3$. Solution: $(3, 2)$.

Time management: identifying quick wins vs. time sinks

5 Compare the effort needed to solve $x + 5 = 12$ versus $3x^2 - 2x - 8 = 0$. Explain a time-management strategy for approaching a timed test section with a mix of question difficulties.

$x + 5 = 12$ requires only one step ($x = 7$) and should take seconds. $3x^2 - 2x - 8 = 0$ requires factoring or the quadratic formula (factors as $(3x + 4)(x - 2) = 0$, giving $x = -\frac{4}{3}$ or $x = 2$) and takes noticeably longer. A sound strategy: scan the section first, answer the quick, straightforward questions immediately to bank easy points, then return to the more time-consuming problems with the remaining time, avoiding getting stuck early on a hard question at the expense of easier ones later in the section.

Using calculator tables to explore function behavior

6 A function is given by $f(x) = x^2 - 6x + 5$. Using a table of values at $x = 0, 1, 2, 3, 4, 5, 6$, identify the zeros of the function and the approximate location of its minimum.

Computing values: $f(0) = 5$, $f(1) = 0$, $f(2) = -3$, $f(3) = -4$, $f(4) = -3$, $f(5) = 0$, $f(6) = 5$. The zeros occur at $x = 1$ and $x = 5$ (where $f(x) = 0$). The minimum value appears at $x = 3$, where $f(3) = -4$ is the smallest value in the table — consistent with the vertex of a parabola lying midway between its two zeros: $\frac{1 + 5}{2} = 3$.

Plugging in answer choices strategically

7 A multiple-choice question asks to solve $2x^2 - x - 6 = 0$, with choices A) $x = -1.5, 2$, B) $x = 1.5, -2$, C) $x = 3, -2$, D) $x = -1.5, -2$. Test choice C by substitution to determine if it's correct.

Testing $x = 3$: $2(9) - 3 - 6 = 18 - 3 - 6 = 9 \neq 0$ — fails, so $x = 3$ is not a root, eliminating choice C. (For reference, factoring $2x^2 - x - 6 = (2x + 3)(x - 2)$ gives the actual roots $x = -1.5$ and $x = 2$, matching choice A.)

Substituting numbers for variables in abstract problems

8 A problem states: if a is a positive integer, which expression is always even: $2a + 1$, a^2, or $4a$? Test with $a = 1$ and $a = 2$ to determine the answer.

For $a = 1$: $2(1) + 1 = 3$ (odd), $1^2 = 1$ (odd), $4(1) = 4$ (even). For $a = 2$: $2(2) + 1 = 5$ (odd), $2^2 = 4$ (even), $4(2) = 8$ (even). Since $4a$ is even in both test cases (and algebraically, $4a$ is always a multiple of 4, hence always even), the answer is $4a$ — testing with simple substituted values quickly reveals the pattern without needing a general proof.

Working backward from answer choices in word problems

9 A word problem states that a number increased by 40% equals 84, with answer choices A) 50, B) 60, C) 70, D) 84. Use the answer choices to work backward and find the correct original number.

Testing choice B (60): $60 \times 1.40 = 84$ ✓. This matches exactly, confirming the original number is 60. Working backward from answer choices is efficient here because it avoids setting up and solving the equation $x(1.4) = 84$ directly, though solving directly gives the same result: $x = \frac{84}{1.4} = 60$.

Deciding when a graph is faster than algebra

10 A question asks for the number of solutions to $|x - 3| = 2x + 1$ without asking for the actual solution values. Explain how graphing $y = |x - 3|$ and $y = 2x + 1$ answers this faster than solving algebraically, then confirm algebraically.

Graphing both functions shows immediately how many times the V-shaped absolute value graph crosses the line — counting intersection points answers a "how many solutions" question without needing exact values, saving time. Algebraically: Case 1 ($x \geq 3$): $x - 3 = 2x + 1$, giving $x = -4$, but this contradicts $x \geq 3$, so reject. Case 2 ($x < 3$): $-(x - 3) = 2x + 1$, so $3 - x = 2x + 1$, giving $3x = 2$, $x = \frac{2}{3}$, which is valid since $\frac{2}{3} < 3$. There is exactly one solution, confirming what the graph would show as a single intersection point.

Recognizing when an exact answer is required vs. an estimate

11 A question asks for the exact value of $\sqrt{72}$ in simplest radical form, while a related question asks for a decimal approximation to the nearest tenth. Provide both, and explain why a calculator alone is insufficient for the first part.

Exact simplified form: $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$. Decimal approximation: $6\sqrt{2} \approx 6(1.414) \approx 8.5$. A calculator alone is insufficient for the exact form because it only returns a decimal approximation ($8.485...$) — recognizing the simplest radical form requires identifying the largest perfect-square factor of 72 algebraically, a step a calculator does not perform automatically.

A multi-part strategy problem: combining estimation, elimination, and verification

12 This question has three parts. A word problem states: "A rectangular garden has a perimeter of 46 feet and an area of 120 square feet. Find its dimensions." Answer choices are A) 10 by 12, B) 8 by 15, C) 6 by 17, D) 5 by 18. (a) Use the perimeter condition alone to eliminate choices quickly by estimation. (b) Check the remaining choices against the area condition. (c) State the correct dimensions and verify both conditions algebraically.

(a) Perimeter $= 2(l + w) = 46$ means $l + w = 23$. Checking each choice's sum: A) $10 + 12 = 22$ (close but not 23), B) $8 + 15 = 23$ ✓, C) $6 + 17 = 23$ ✓, D) $5 + 18 = 23$ ✓. This quickly eliminates A. (b) Checking area for the remaining choices: B) $8 \times 15 = 120$ ✓, C) $6 \times 17 = 102$ (too small), D) $5 \times 18 = 90$ (too small). Only B satisfies both conditions. (c) Dimensions: 8 by 15. Verify: perimeter $= 2(8 + 15) = 2(23) = 46$ ✓; area $= 8 \times 15 = 120$ ✓.

Avoiding common sign and order-of-operations errors

13 A student evaluates -3^2 and gets 9, but the correct answer is -9. Explain the order-of-operations error, then correctly evaluate -3^2 and $(-3)^2$ to show the difference.

The error is treating the negative sign as part of the base being squared. By the order of operations, exponentiation applies before the unary negative sign is applied: -3^2 means $-(3^2) = -9$, NOT $(-3)^2$. In contrast, $(-3)^2$ explicitly squares the negative number: $(-3)^2 = (-3) \times (-3) = 9$. The parentheses make all the difference — always check whether a negative sign is inside or outside the exponentiation.